

Constructing Optimal Rank-Decomposition Trees of Biconnected Outerplanar Graphs

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Abstract

A graph G is called an outerplanar graph if it has an embedding in the plane such that every vertex lies on the boundary of the exterior face. In this paper, we show that the rankwidth of a biconnected outerplanar graph is at most two. Further, we propose a linear-time algorithm to construct an optimal rank-decomposition tree for a biconnected outerplanar graph.

1 Introduction

Let $G = (V, E)$ be a finite, simple, and undirected graph where V and E are the vertex and edge sets of G , respectively. Robertson and Seymour introduced the concept of treewidth on graphs [10, 11]. It has been shown that if the input graphs have bounded treewidths, then many NP-hard problems can be solved in polynomial time. However, large cliques are excluded by graphs of bounded treewidth. Courcelle and Olariu introduced the concept of cliquewidth to exclude the weakness of treewidth [2]. Unfortunately, Fellows, Rosamond, Rotics, and Szeider [4, 5] have shown that the problem of deciding whether a graph has cliquewidth at most k is NP-complete if k is given as an input. Instead, Oum and Seymour proposed the concept of rankwidth [9].

A *cubic* (respectively, *subcubic*) tree is a tree such that every internal vertex has degree 3 (respectively, at most 3). A *rank-decomposition* of a graph G is a pair (T, L) , where T is a cubic tree and $L : V \rightarrow \{v \mid v \text{ is a leaf of } T\}$ is a bijective function. In the case that $|V| = 2$, T contains only one edge. However, there is no rank-decomposition of G if $|V| = 1$. For an edge e of T , the connected components of $T - e$ induce a partition (X, Y) of the set of leaves of T . Since L

is bijection, (X, Y) is also a bipartition of V . Let $X = \{u_1, \dots, u_r\}$ and $Y = \{v_1, \dots, v_s\}$. Let M be an $|X| \times |Y|$ 0-1 matrix such that $M[i, j] = 1$ if $(u_i, v_j) \in E$; otherwise, $M[i, j] = 0$. The *width* of edge e of a rank-decomposition (T, L) is the rank of the matrix M . The *width* of (T, L) is the maximum width of all edges of T . The *rankwidth* $rdw(G)$ of G is the minimum of the width of all rank-decompositions of G . In the case of $|V| = 1$, $rdw(G) = 0$. Note that T is also called a *rank-decomposition tree* of G .

Let $cwd(G)$ (respectively, $twd(G)$) denote the cliquewidth (respectively, treewidth) of G . It is shown that $rdw(G) \leq cwd(G) \leq 2^{rdw(G)+1} - 1$ for any graph G [9]. It implies that $cwd(G)$ can be approximated by $rdw(G)$. That is, cliquewidth is equivalent to rankwidth in the sense that $cwd(G)$ is bounded. Let k be fixed. There is an $O(n^3)$ -time algorithm that either confirms that an n -vertex input graph has rankwidth greater than k or outputs a rank-decomposition of width at most $3k + 1$ [8]. It is also known that $rdw(G) \leq twd(G) + 1$. Thus it is interesting to know whether optimal rank-decompositions of the graph classes of bounded treewidth can be constructed in time lesser than $O(n^3)$. This motivates us to study this problem on outerplanar graphs. Since the treewidth of an outerplanar graph G is at most 2, the rankwidth of G can be at most 3. In this paper, we show that the rankwidth of a biconnected outerplanar graph G is at most 2. First, we propose a linear-time algorithm to construct an optimal rank-decomposition tree T of a maximal outerplanar graph G . We show that the rankwidth of T is at most 2. Then, we show that the rankwidth of a biconnected outerplanar graph G does not greater than the rankwidth of a maximal planar graph that contains G as a subgraph. Finally, we propose a linear-time algorithm to triangulate a biconnected outerplanar graph into a maximal one. Since the class of graphs with rankwidth one is ex-

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actly the class of distance-hereditary graphs, the rank-decomposition tree of a biconnected outerplanar graph computed by our algorithm is optimal.

2 Preliminaries

A graph is called an *outerplanar* graph if it has an embedding in the plane such that every vertex lies on the boundary of the exterior face [3]. If an outerplanar graph is biconnected, then its vertices form a Hamiltonian cycle. That is, all its vertices can be ordered in increasing order according to a clockwise ordering on the cycle. An outerplanar graph is *maximal* if it is not possible to add an edge such that the resulting graph is still outerplanar. That is, every face of a maximal outerplanar graph contains exactly three edges except the exterior face. Of course, a maximal outerplanar graph is biconnected. The class of (maximal) outerplanar graphs can be recognized in linear time [6]. Figure 1 shows examples.

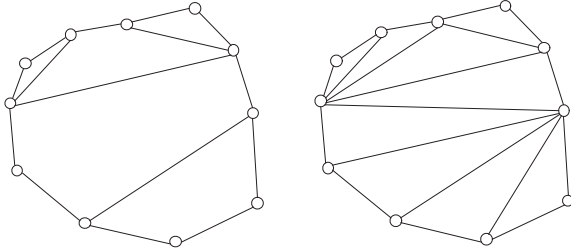


Figure 1: A biconnected outerplanar graph (left) and a maximal outerplanar graph.

3 Maximal outerplanar graphs

In this section, we consider the maximal outerplanar graphs. Let $G = (V, E)$ be a maximal outerplanar graph and let $n = |V|$. It is not hard to check that G has $n - 2$ interior faces. Therefore, $|E| = 2n - 3$. Let f_i denote interior face i for $i \in \{1, \dots, n - 2\}$. We say that f_i contains vertex v if v is on the boundary of f_i . The *skeleton* of G is a graph $S = (F, H)$ where $F = \{f_1, \dots, f_{n-2}\}$ and $H = \{(f_i, f_j) \mid f_i \text{ and } f_j \text{ share the same edge in } G\}$. Since G is planar, $|E|$ and $|F|$ are bounded by $O(n)$. Thus, S can be constructed in $O(n)$ time. Note that since G is maximal outerplanar, the boundary of each

interior face contains exactly three edges. Thus the degree of a vertex in S is at most 3. That is, S is a subcubic tree.

Without loss of generality, we assume that the vertices are indexed by a clockwise order. For convenience, f_i denotes a face in G and also denotes a vertex in S if there is no confusion. By using the skeleton of G , we visit the vertices one by one according to their indices. Our algorithm is presented as follows.

Algorithm 1 buildtree(a maximal outerplanar graph G)

- 1: Let $G = (V, E)$ with $V = \{v_1, \dots, v_n\}$;
 - 2: Construct the skeleton $S = (F, H)$ of G ;
 - 3: Let $T = (F \cup V, H)$;
 - 4: **for** $i = 1$ to n **do**
 - 5: Let f_j be the face containing vertices v_i and v_{i+1} ;
 - 6: $H = H \cup \{(v_i, f_j)\}$;
 - 7: **end for**
 - 8: **return** T ;
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It is easy to check that Algorithm 1 runs in $O(n)$ time. We now prove that our algorithm can correctly construct an optimal rank-decomposition tree of G . We first show that T , the output of our algorithm, is a cubic tree.

Lemma 1 Let $G = (V, E)$ be a maximal outerplanar graph with $n = |V| \geq 3$. Then, the tree T constructed by Algorithm 1 is a cubic tree.

Proof. Let $S = (F, H)$ be the skeleton of G . First, $|F| = n - 2$. The total degree of S is $2 * (|F| - 1) = 2n - 6$. Since there are n vertices linking to S in T , the total degree of the internal vertices of T is $(2n - 6) + n = 3n - 6$. Since T has exactly $|F| = n - 2$ internal vertices, the degree of each internal vertex of T is 3. Therefore, T is a cubic tree. $\square$

We now check the rankwidth produced by T .

Lemma 2 Let $G = (V, E)$ be a maximal outerplanar graph with $|V| > 3$ and T be the cubic tree constructed from G by Algorithm 1. Then the rankwidth determined by T is 2.

Proof. Consider any edge e of T . That is, e belongs to the skeleton of G . It is easy to check that if e is the edge incident from a leaf of T , then its corresponding rank is 1. Thus we consider e being the edge incident from two internal vertices

of T . Further, we assume that e is constructed by the edge $(a, b) \in E$ during the construction of the skeleton. Let A and B be the partition of V by the edge e . Figure 2 shows a possible partition. Let $a, c \in A$ and $b, d \in B$ as shown in Figure 2. Then we obtain a rank corresponding to e as follows:

$$\text{rank of } \begin{pmatrix} & a & \cdots & c \\ b & 1 & * & \cdots & * & 1 \\ & * & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & & \\ & * & 0 & & & \\ d & 1 & 0 & \cdots & 0 & 0 \end{pmatrix} = 2$$

We notice that the symbol “*” denotes “don’t care”, *i.e.*, it can be either 1 or 0. Since e is determined by the edge (a, b) and G is a maximal outerplanar graph, vertices in A (respectively, B) can be only adjacent to b (respectively, a). That is, the rankwidth determined by T is 2. $\square$

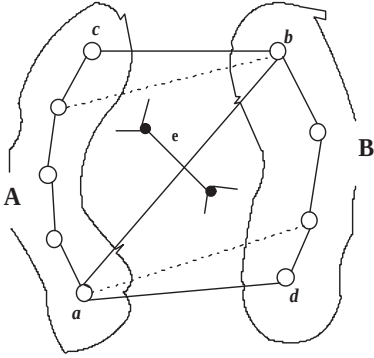


Figure 2: A partition by the edge e .

A graph G is *distance hereditary* if the distance of any two vertices remains the same in every induced connected subgraph of G [1]. Oum proved that the graphs of rankwidth 1 are exactly the distance-hereditary graphs [7].

Theorem 1 *The rankwidth of the class of maximal outerplanar graphs is at most 2.*

Proof. The upper bound is from Lemma 2. Since the triangle is both maximal outerplanar and distance hereditary, the theorem holds. $\square$

Corollary 1 *Algorithm 1 produces an optimal rank-decomposition tree of a maximal outerplanar graph in linear time.*

4 Biconnected outerplanar graphs

In this section, we consider the class of biconnected outerplanar graphs. A *triangulation* of a biconnected outerplanar graph G is a biconnected outerplanar graph G' such that the boundary of each interior face of G' is a cycle of length 3. That is, a triangulation of a biconnected outerplanar graph is a maximal outerplanar graph. Let $G = (V, E)$ be any graph. For an edge $e \in E$, let $G - e$ be the subgraph $G' = (V, E \setminus \{e\})$. For a vertex $v \in V$, let $G - v$ be the subgraph induced by $V \setminus \{v\}$. It is easy to see that the rankwidth of $G - v$ is smaller or equal to the rankwidth of G . However, the rankwidth of $G - e$ is not necessary smaller or equal to the rankwidth of G . For example, the rankwidth of K_4 , *i.e.*, a complete graph with four vertices, is 1. However, the rankwidth of $K_4 - e$ for any edge e is 2 since $K_4 - e$ is a maximal outerplanar graph with four vertices. Thus it is not clear that the rankwidth of a biconnected outerplanar graph is smaller, greater, or equal to the rankwidth of its triangulation. In the following, we will show that the rankwidth of a biconnected outerplanar graph is no more than the rankwidth of its triangulation.

Let us first consider a triangulation G' of a biconnected outerplanar graph G . Since G' is a maximal outerplanar graph, we obtain an optimal rank-decomposition tree T of rankwidth 2 for G' . We claim that T is also an optimal rank-decomposition tree for G . Let us consider an internal edge e of T . Figure 3 shows a possible partition of V . Note that (a, b) may be not an edge in G .

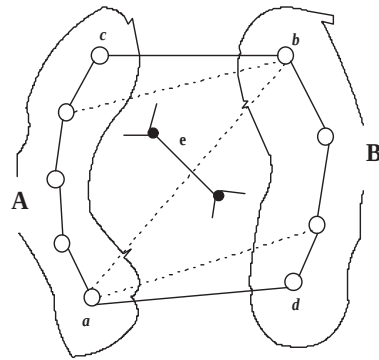


Figure 3: A partition by the edge e (e may be not in G).

Therefore, we obtain the rank corresponding to

e as follows:

$$\text{rank of } \begin{pmatrix} & a & \cdots & c \\ b & * & * & \cdots & * & 1 \\ & * & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & & \\ & * & 0 & & & \\ d & 1 & 0 & \cdots & 0 & 0 \end{pmatrix} = 2$$

That is, the rankwidth of G is not greater than the rankwidth of its triangulation. By a similar argument for maximal outerplanar graphs, we have the following theorem.

Theorem 2 *The rankwidth of the class of biconnected outerplanar graphs is at most 2.*

The last job we should do is to transform a biconnected outerplanar graph into a maximal outerplanar graph. The idea of our triangulating algorithm is to process vertices one by one according to their indices. For each stage, we detect a cycle C_k , $k > 3$, and then triangulate it. Let $N(v) = \{u \mid (u, v) \in E\}$. For a vertex set $W \subseteq V$, let $\max(W)$ denote the vertex with the maximum index in W . Let $\text{Dequeue}(Q)$ be the function that returns the first element of queue Q and removes it from Q . Let $\text{Enqueue}(Q, v)$ be the operation that adds the element v into the rear of queue Q . The detail of our algorithm is as follows.

Algorithm 2 triangulation(a biconnected outerplanar graph $G = (V, E)$)

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1: Let  $V = \{v_1, \dots, v_n\}$  and  $Q = \emptyset$  be an empty
   set;
2:  $\text{Enqueue}(Q, v_1)$ ;
3: while  $Q \neq \emptyset$  do
4:    $v_i = \text{Dequeue}(Q)$ ;
5:    $v_k = \max(N(v_i))$ ;
6:    $v_r = \max(N(v_{i+1}))$ ;
7:   if  $v_r > v_{i+3}$  then
8:      $\text{Enqueue}(Q, v_{i+1})$ ;
9:   end if
10:  while  $v_r \neq v_k$  do
11:     $E = E \cup \{(v_i, v_r)\}$ ;
12:    if  $v_{r+2} \leq v_k$  and  $\max(N(v_r)) > v_{r+2}$ 
       then
13:       $\text{Enqueue}(Q, v_r)$ ;
14:    end if
15:     $v_r = \max(N(v_r))$ ;
16:  end while
17: end while
18: return  $G$ ;
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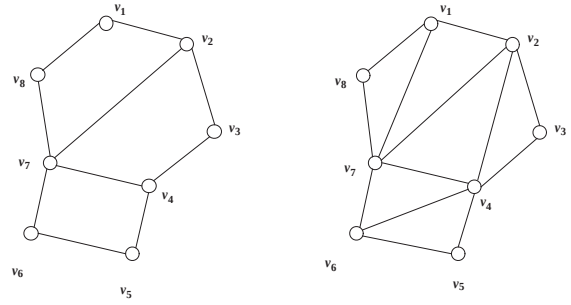


Figure 4: Triangulating a biconnected outerplanar graph.

Figure 4 shows an example for a triangulation of a biconnected outerplanar graph.

Since G is planar, it is easy to check that Algorithm 2 runs in $O(n)$ time. In general, an outerplanar graph is not necessary biconnected. After obtaining a rank-decomposition tree for each biconnected component, we can then concatenate them into a rank-decomposition tree of G .

5 Conclusion

In this paper we show that the rankwidth of a biconnected outerplanar graph is at most two, and a corresponding rank-decomposition tree can be found in linear time. It is interesting to construct optimal rank-decomposition trees for other classes of graphs.

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